Class Field Theory

1 0908

Goal Describe all abelian extension of number fields.
i.e. inverse Galois problem for abelian groups.

Recall (Kronecker-Weber Theorem) Every abelian extension of the rational numbers is contained in a cyclotomic field.

What happens if we replace $\mathbb{Q}$ by a number field K?

Denote $\mathrm{Cl}(K)$ as the ideal class group of K. (Fractional ideals modulo principal ideals.)

Theorem 1.1 : Let L be the maximal unramified abelian extension of a number field K. Then L/K is finite and $\mathrm{Gal}(L/K) \cong \mathrm{Cl}(K)$.
L is called the Hilbert class field of K.

Recall Let $N > 1$ be an integer. ζ_N denotes a primitive N-th root of unity. Then

1. $\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \cong (\mathbb{Z}/N\mathbb{Z})^\times$.
2. Suppose $N \not\equiv 2 (\mathrm{mod}\, 4)$. Then p is ramified in $\mathbb{Q}(\zeta_N)$ if and only if $p \mid N$.
3. If p splits completely in $\mathbb{Q}(\zeta_N)$, then p is unramified, so the decomposition group is isomorphic to $\mathrm{Gal}(\kappa(P)/\kappa(p))$.
   The Frobenius element $\mathrm{Frob}(P/p)$ is the unique element in the decomposition group that acts as $x \mapsto x^p$ on the residue field.

It is too restrictive to ask for unramified extensions.
So we ask for abelian extensions.

1.1 Review $\mathbb{A}_K, \mathbb{I}_K = \mathbb{A}_K^\times, \mathrm{Cl}(K)$

Let I be a fractional ideal in K. Then I can be written as $I = P_1^{a_1} \ldots P_r^{a_r}$, where P_i are prime ideals and $a_i \in \mathbb{Z}$.
So the group of fractional ideals is a free abelian group generated by the prime ideals.

$$\bigoplus_P \mathbb{Z}$$

and Cl is the quotient of $\bigoplus_P \mathbb{Z}$ by the image of the map $K^\times \to \bigoplus_P \mathbb{Z}$ given by $x \mapsto (v_P(x))_P$, where v_P is the valuation at P.

Let $U = \prod_{v:\mathrm{inf}} K_v^\times \times \prod_{v:\mathrm{fin}} \mathcal{O}_v^\times$. It is an open subgroup of $\mathbb{A}_K^\times$.
Then $\mathbb{A}_K^\times / U$ is exactly $\bigoplus_{v:\mathrm{fin}} \mathbb{Z}$. So we have a surjective map $\mathbb{A}_K^\times \to \mathrm{Cl}(K)$, and the kernel is $UK^\times$.

As the class group corresponds to the maximal unramified abelian extension, we want to shrink U to a smaller open subgroup to detect more abelian extensions.

[Ray class group]@Obsidian

Remark: This isomorphism shows that $\mathrm{Cl}(K, \mathfrak{a})$ is finite:
By the isomorphism, it is discrete.
Also, it is a quotient of $\mathbb{A}_K^1/K^\times$, which is compact, so it is finite.

Example 1.2 : $\mathrm{Cl}(\mathbb{Q}, N) \cong (\mathbb{Z}/N\mathbb{Z})^\times$.

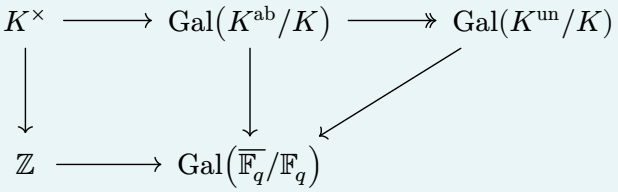
[計算 Cl(Q,N)]@Obsidian

By Kronecker-Weber, we know that $\mathrm{Gal}\big(\mathbb{Q}^{\mathrm{ab}}/\mathbb{Q}\big)$ is the inverse limit of $(\mathbb{Z}/N\mathbb{Z})^\times$ over N.
By above, $(\mathbb{Z}/N\mathbb{Z})^\times$ is a quotient of $C_{\mathbb{Q}} = \mathbb{A}_{\mathbb{Q}}^\times/\mathbb{Q}^\times$.
Class field theory is saying that we can construct a map from $C_K = \mathbb{A}_K^\times/K^\times$ to $\mathrm{Gal}\big(K^{\mathrm{ab}}/K\big)$ for any number field K, and the image of this map is dense.

Theorem 1.3 (Local class field theory): Let K be a local field. There is a unique continuous homomorphism

$$\rho_K : K^\times \to \mathrm{Gal}\big(K^{\mathrm{ab}}/K\big)$$

satisfying the following properties:

1. For any L/K finite abelian extension, the kernel of ρ_K is exactly $N_{L/K}(L^\times)$, the norm of $L^\times$ in $K^\times$.
2. 

3. There is a 1-1 correspondence between U: an open subgroup of $\mathrm{Gal}\big(K^{\mathrm{ab}}/K\big)$ and V: open subgroup of finite index of $K^\times$ given by $U \mapsto \rho_K^{-1}(U)$.

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Definition (CSA): A (finite-dimensional) central simple algebra (CSA) over a field K is a finite-dimensional K-algebra R such that:

- R is a finite-dimensional K-algebra.
- R is central: The center of R is exactly K.
- R is simple: R has no nontrivial two-sided ideals.

Example 2.1 : The matrix algebra $M_{n(K)}$ is a CSA over K.

Fact 2.2 (Wedderburn): Let A be a CSA over K. Then there's a central division algebra D s.t.

$$A \cong M_n(D)$$

for some $n \in \mathbb{Z}_{>0}$ and D is unique up to isomorphism.

Definition Let A, B be two CSAs over K. We say A and B are Brauer equivalent if there exist $m, n \in \mathbb{Z}_{>0}$ s.t.

$$A \otimes_K M_m(K) \cong B \otimes_K M_n(K)$$

i.e. their corresponding division algebras are isomorphic. The Brauer equivalence class of A is denoted by $[A]$.
The Brauer group $\mathrm{Br}(K)$ is the set of Brauer equivalence classes of CSAs over K with the group operation given by tensor product over K.
Then $[A]^{-1} = [A^{\mathrm{op}}]$.

The importance of the Brauer group in CFT:
If K_v is non-archimedean, then $\mathrm{Br}(K_v) \cong \mathbb{Q}/\mathbb{Z}$ and it is canonical.

We can write $\mathrm{Br}(K_v)$ explicitly as its element is the form (χ, a) where χ is a character of $\mathrm{Gal}\big(K_v^{\mathrm{ab}}/K_v\big)$ and $a \in K_v^\times$.
It help us define a map

$$\begin{aligned} \rho : K_v^\times &\to \mathrm{Hom}\big(\mathrm{Hom}\big(\mathrm{Gal}\big(K_v^{\mathrm{ab}}/K_v\big), \mathbb{Q}/\mathbb{Z}\big), \mathbb{Q}/\mathbb{Z}\big) \\ a &\mapsto (\chi \mapsto \mathrm{inv}(a, \chi)) \end{aligned}$$

and the right hand side is $\mathrm{Gal}\big(K_v^{\mathrm{ab}}/K_v\big)$ given by Pontryagin duality.

3 Lecture 1: Group cohomology

Let G be a finite group.

Definition

1. M is a G-module if M is an abelian group with a G-action that is compatible with the group structure.
   We also consider M as a $\mathbb{Z}[G]$-module.
2. $f : M \to N$ is a G-module homomorphism if f is a map preserving the G-action.

Definition $C^n(\mathbb{Z}[G], A) = \mathrm{Hom}_{G(\mathbb{Z}[G^{n+1}], A)}$.
$\partial^n : C^n(\mathbb{Z}[G], A) \to C^{n+1}(\mathbb{Z}[G], A)$ is defined by

$$(\partial^n f)(g_0, ..., g_{n+1}) = \sum_{i=0}^{n+1} (-1)^i f(g_0, ..., \hat{g_i}, ..., g_{n+1})$$

It defines a cochain complex $(C^n(\mathbb{Z}[G], A), \partial^n)$, and the cohomology of this complex is denoted by $H^n(G, A)$.

4 0917

Definition A G-module B is called relatively injective if B is a direct summand of an induced module $\mathrm{Ind}_1^G(A)$ for some abelian group A.

Corollary 4.1 : A relatively injective G-module is acyclic.

4.1 Change of groups

General: $f : G' \to G$: group homomorphism of finite groups.
Let A be a G-module. Through f, we can make A a G'-module. Denote as f^*A. This is a functor $f^* : G\text{-Mod} \to G'\text{-Mod}$. We have a natural map $f^* : H^n(G, A) \to H^n(G', f^*A)$ for all n.

Now consider the case $H \hookrightarrow G$. The pullback map is called the restriction map.

Theorem 4.2 (Shapiro lemma):

$$H^n(H, A) \cong H^n\big(G, \mathrm{Ind}_H^G(A)\big)$$

Proof: By Frobenius reciprocity, we have

$$\mathrm{Hom}_G\big(B, \mathrm{Ind}_H^G(A)\big) \cong \mathrm{Hom}_H(B, A)$$

For $n = 0$: Take $B = \mathbb{Z}$ with trivial action.
Now for $n > 0$, take a injective resolution $A \to I^\bullet$. By Frobenius reciprocity and the exactness of induction functor, $\mathrm{Ind}_H^G(I^\bullet)$ is an injective resolution of $\mathrm{Ind}_H^G(A)$.
Applying the case $n = 0$ to $I^\bullet$, we get the desired isomorphism since both sides are universal δ-functors. ■

Consider another case $H \hookrightarrow G$ and A is a G-module. Consider the following map:
For $n = 0$, we have a map $H^0(H, A|_H) \to H^0(G, A)$ given by $a \mapsto \sum_{g \in G/H} ga$.
Now, $H^n(H, A|_H)$ is a δ-functor. We want to show that it's the derived functor of $A \mapsto A^H = H^0(H, A|_H)$.
By the dual of Frobenius reciprocity,

$$\mathrm{Hom}_H(- \otimes_H G, I) \cong \mathrm{Hom}_G(-, I)$$

Since $- \otimes_H G$ is exact, $I^\bullet|_H$ is an injective resolution of $A|_H$. (Restriction is exact.)
This proves that $H^n(H, A|_H)$ is the derived functor of $A \mapsto A^H$ hence a universal δ-functor.
As $A \mapsto H^n(G, A)$ is also a δ-functor, we have a natural transformation $H^n(H, A|_H) \to H^n(G, A)$ for all n.

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5.1 Inflation

Let $H \lhd G$ be a normal subgroup.
Given a G-module A, we can make A^H a G/H-module.
Then we have a natural map

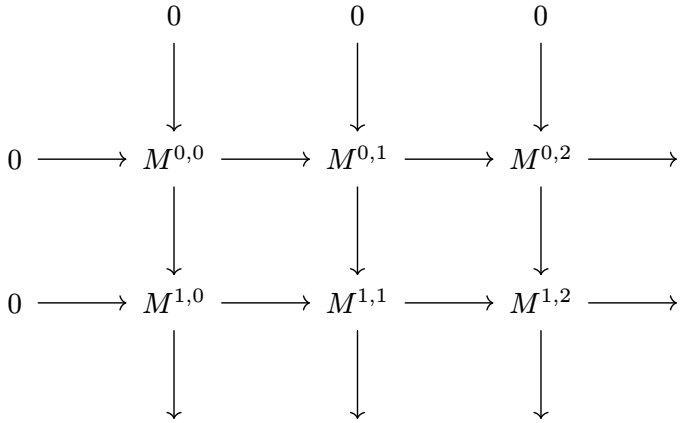
$$\inf_{G/H} : H^n(G/H, A^H) \to H^n(G, A^H) \to H^n(G, A)$$

Theorem 5.1 : We have an exact sequence

$$0 \longrightarrow H^1(G/H, A^H) \xrightarrow{\inf_{G/H}} H^1(G, A) \xrightarrow{\text{res}} H^1(H, A)$$

5.2 Intermezzo: Spectral sequences (arising from a double complex)

Set-up: $M^{\bullet,\bullet}$: bounded below double complex of abelian groups.



with

- $d_\to \circ d_\to = 0$
- $d_\downarrow \circ d_\downarrow = 0$
- $d_\to \circ d_\downarrow + d_\downarrow \circ d_\to = 0$

Define the total complex $M^\bullet$ by $M^n = \sum_{p+q=n} M^{p,q}$ and $d = d_\to + d_\downarrow$.
We want to compute the cohomology of $M^\bullet$.
The strategy is to filter $M^\bullet$ by p and q and get two spectral sequences.

Define $F^i M^n = \sum_{p \geq i} M^{p,n-p}$.
Then

$$F^i H^n = \left(Z^n \cap F^i M^n\right) + B^n / B^n \subset H^n$$

It gives a filtration of H^n:

$$0 \subset F^n H^n \subset F^{n-1} H^n \subset \ldots \subset F^0 H^n = H^n$$

- Page 0: $E_0^{p,q} = M^{p,q}$
- Page 1: $E_1^{p,q} = H^q(M^{p,\bullet})$ with the differential $d_1^{p,q} : E_1^{p,q} \to E_1^{p+1,q}$ induced by $d_\downarrow$.
- Page 2: $E_2^{p,q} = H^p(E_1^{\bullet,q})$ with the differential $d_2^{p,q} : E_2^{p,q} \to E_2^{p+2,q-1}$ induced by $d_\downarrow$:

eye The map is given by:
For $[x] \in E_2^{p,q}$, we have $x \in M^{p,q}$ s.t. $d_\rightarrow x = 0$ and $d_\downarrow x = d_\rightarrow(-y)$, for some $y \in M^{p+1,q-1}$. Then $d_2^{p,q}([x]) = \left[d_\downarrow^{p+1,q-1}(y)\right]$.

Inductively, we can define

$$E_{r+1}^{p,q} = \ker(d_r^{p,q}) / \operatorname{im}(d_r^{p-r,q+r-1})$$

and

$$d_r^{p,q} : E_r^{p,q} \to E_r^{p+r,q-r+1}$$

For a fixed (p,q), the sequence $E_r^{p,q}$ stabilizes for r large enough. Denote the stable value by $E_\infty^{p,q}$.
If for some $x \in M^{p,q}$, we have $d_\rightarrow x = 0$ and $d_\downarrow x = 0$, i.e. $dx = 0$. Then $d_n(x) = 0$ for all n, so $[x] \in E_\infty^{p,q}$.
Similarly, if $x \in M^{p,q}, y \in M^{p+1,q-1}$ and $d(x+y) = 0$, then $[x+y] \in E_\infty^{p,q}$.
In fact,

$$E_\infty^{p,q} = \frac{Z^n \cap F^p M^{p+q}}{Z^n \cap F^{p+1} M^{p+q} + B^n \cap F^p M^{p+q}}$$

which is exactly the p-th graded piece of the filtration of H^n i.e. $\frac{F^p H^n}{F^{p+1} H^n}$.

Now, for lower degree terms, we have
- $F^1 H^1 = E_\infty^{1,0}$
- $H^1 / F^1 H^1 = E_\infty^{0,1}$
- $F^2 H^2 = E_\infty^{2,0}$

i.e.

$$0 \longrightarrow E_\infty^{1,0} \longrightarrow H^1 \longrightarrow E_\infty^{0,1} \longrightarrow 0$$

We have $E_\infty^{1,0} = E_2^{1,0}$ and $E_\infty^{0,1} = \ker\left(d_2^{0,1}\right)$.
Also, $E_\infty^{2,0} = \operatorname{coker}\left(d_2^{0,1}\right)$.
So we have an exact sequence

$$0 \longrightarrow E_2^{1,0} \longrightarrow H^1 \longrightarrow E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \longrightarrow H^2$$

Theorem 5.2 (Grothendieck spectral sequence): Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be three abelian categories.
$F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{B} \to \mathcal{C}$ be two additive left-exact functors.
Assume that

- $\mathcal{A}, \mathcal{B}$ have enough injectives.
- F sends injective objects of $\mathcal{A}$ to G-acyclic objects of $\mathcal{B}$.

Then

$$E_2^{p,q} = (R^pG)(R^qF)(A) \Longrightarrow R^{p+q}(G \circ F)(A)$$

For $A \in \mathcal{A}$, take an injective resolution $A \to I^\bullet$.
Then $F(I^\bullet)$ is a complex in $\mathcal{B}$ of acyclic objects.

Fact 5.3 : There is a double complex $J^{p,q}$ s.t. $J^{p,\bullet}$ is an injective resolution of $F(I^p)$ for all p.

Example 5.4 :

- $\mathcal{A} = \mathrm{Mod}_G, \mathcal{B} = \mathrm{Mod}_{G/H}, \mathcal{C} = \mathrm{Ab}$.
- $F(A) = A^H, G(B) = B^{G/H}$.

Then $R^pG = H^p(G/H, -)$, $R^qF = H^q(H, -)$, and $R^{p+q}(G \circ F) = H^{p+q}(G, -)$.
So we have a spectral sequence

$$0 \longrightarrow E_2^{1,0} \longrightarrow H^1 \longrightarrow E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \longrightarrow H^2$$

which can be rewritten as

$$0 \longrightarrow H^1(G/H, A^H) \xrightarrow{\inf_{G/H}} H^1(G, A) \xrightarrow{\mathrm{res}} H^1(H, A)^{G/H} \xrightarrow{d_2} H^2(G/H, A^H) \longrightarrow H^2(G, A)$$

5.3 Application

Let G be a finite group and A be a G-module.

Theorem 5.5 : If $H \leq G$ is a subgroup of index m, then

$$\mathrm{cor}_G^H \circ \mathrm{res}_G^H = m \, \mathrm{id}_{H^n(G,A)}$$

Proof: For $n = 0$, it is clear.
By dimension shifting, we are done. ∎

Corollary 5.6 : Take $H = 1$, then we know that $H^n(G, A)$ is of $|G|$-torsion for $n > 0$.

Corollary 5.7 : If A is of finite type, then $H^n(G, A)$ is finite for $n > 0$.

Proof: By the previous corollary, $H^n(G, A)$ is of $|G|$-torsion.
Also, $H^n(G, A)$ is finitely generated abelian group since A is finitely generated by the cochain complex construction.
So it is finite. ∎

Corollary 5.8 : Suppose that A is uniquely divisible i.e. A is a $\mathbb{Q}$-vector space. Then $H^n(G, A) = 0$ for $n > 0$.

Proof: Multiplication by $|G|$ is an automorphism of A since A is uniquely divisible.
But it is zero map on $H^n(G, A)$ for $n > 0$. So $H^n(G, A) = 0$ for $n > 0$. ∎

Example 5.9 : Consider the exact sequence of G-modules

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

with the trivial G-action. Then by the long exact sequence of cohomology, we have

$$H^1(G, \mathbb{Q}/\mathbb{Z}) \cong H^2(G, \mathbb{Z})$$

Now suppose G is a Galois group of a finite Galois extension L/K.
Since $\mathbb{Q}/\mathbb{Z}$ is of trivial G-action, we have

$$H^1(G, \mathbb{Q}/\mathbb{Z}) \cong \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z}) = \mathrm{Hom}\big(G^{\mathrm{ab}}, \mathbb{Q}/\mathbb{Z}\big)$$

As any finite subgroup of $\mathbb{Q}/\mathbb{Z}$ is cyclic, we have

$$G/\ker f \cong \mathrm{im}\, f \leq \mathbb{Q}/\mathbb{Z}$$

and $\mathrm{im}\, f \cong \mathbb{Z}/n\mathbb{Z}$ for some n.
So to give a f is equivalent to giving a cyclic quotient of G and a generator of the cyclic group $\mathrm{Gal}\big(L^{\ker f}/K\big)$.

Example 5.10 : Same as above. We have $H^1(G, \mathbb{Z}) = 0$.

6 Lecture 2: Tate cohomology

Definition Let G be a finite group.

1. The element norm $\alpha_0 \in \mathbb{Z}[G]$ is defined as
$$\alpha_0 = \sum_{g \in G} g$$
2. Let A be a G-module. The norm map $N : A \to A$ is defined as
$$N(a) := \alpha_0 a = \sum_{g \in G} ga$$
3. The augmentation (two-sided) ideal I_G is defined as
$$I_G = \ker(\mathbb{Z}[G] \to \mathbb{Z}) = \mathrm{span}_{\mathbb{Z}}(g - 1 : g \in G)$$
4. $A_G := A/I_G A = \mathbb{Z}[G]/I_G \otimes_{\mathbb{Z}[G]} \otimes A$, which is the maximal quotient of A on which G acts trivially.

As the functor $A \mapsto A_G$ is given by a tensor product, it is right-exact.
So we can define its left derived functor, denote by $H_n(G, A)$.

Example 6.1 : $H_1(G, \mathbb{Z}) = G^{\mathrm{ab}}$.

Proof: Consider the exact sequence of G-modules
$$0 \longrightarrow I_G \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0$$
We have the long exact sequence of homology
$$0 \longrightarrow H_1(G, \mathbb{Z}) \longrightarrow I_G/I_G^2 \longrightarrow \mathbb{Z}[G]_G = \mathbb{Z} \longrightarrow \mathbb{Z}_G = \mathbb{Z} \longrightarrow 0$$
So $H_1(G, \mathbb{Z}) \cong I_G/I_G^2$.
Consider the map $G \to I_G/I_G^2$ given by $g \to [g - 1]$.
Since $I_G = \mathrm{span}_{\mathbb{Z}}(g - 1 : g \in G)$, the map is surjective.
Also, since I_G/I_G^2 is abelian, the map factors through G^{ab}.
Now consider the map $I_G \to G^{\mathrm{ab}}$ given by $g - 1 \to g$.
It is surjective and factors through I_G/I_G^2.

6.1 0929

Definition (Tate cohomology): Let G be a finite group and A be a G-module. The Tate cohomology groups $\hat{H}^n(G, A)$ are defined as follows:

- For $n > 0$, $\hat{H}^n(G, A) = H^n(G, A)$.
- For $n = 0$, $\hat{H}^0(G, A) = A^G / N_G(A)$.
- For $n = -1$, $\hat{H}^{-1}(G, A) = \ker(N_G)/I_G A$.
- For $n < -1$, $\hat{H}^n(G, A) = H_{-n-1}(G, A)$.

where N_G is the norm map and I_G is the augmentation ideal.

Theorem 6.2 : Let $0 \to A \to B \to C \to 0$ be a short exact sequence of G-modules. Then we have a long exact sequence of Tate cohomology groups

$$\begin{array}{ccccccc}
\ldots & \xrightarrow{\delta} & \hat{H}^{-2}(G,A) & \longrightarrow & \hat{H}^{-2}(G,B) & \longrightarrow & \hat{H}^{-2}(G,C) \\
 & & & \overset{\delta}{\swarrow} & & & \\
 & & \hat{H}^{-1}(G,A) & \longrightarrow & \hat{H}^{-1}(G,B) & \longrightarrow & \hat{H}^{-1}(G,C) \\
 & & & \overset{\delta}{\swarrow} & & & \\
 & & \hat{H}^{0}(G,A) & \longrightarrow & \hat{H}^{0}(G,B) & \longrightarrow & \hat{H}^{0}(G,C) \\
 & & & \overset{\delta}{\swarrow} & & & \\
 & & \hat{H}^{1}(G,A) & \longrightarrow & \hat{H}^{1}(G,B) & \longrightarrow & \hat{H}^{1}(G,C) \xrightarrow{\delta} \ldots
\end{array}$$

Proof: We have the following exact sequences of abelian groups:

$$\begin{array}{ccccccccc}
H_1(G,C) & \xrightarrow{\delta} & A_G & \longrightarrow & B_G & \longrightarrow & C_G & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle N} & & \downarrow{\scriptstyle N} & & \downarrow{\scriptstyle N} & & \\
0 & \longrightarrow & A^G & \longrightarrow & B^G & \longrightarrow & C^G & \xrightarrow{\delta} & H^1(G,A)
\end{array}$$

by the snake lemma, we get the desired long exact sequence of Tate cohomology groups. ∎

Similar to the case of group cohomology, we have the following results:
The group coholomogy can be computed by applying the functor $\mathrm{Hom}_{G(-,A)}$ to the exact sequence

$$\ldots \longrightarrow \mathbb{Z}[G^2] \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0$$

and compute the cohomology of the resulting complex.
For the group homology, we can apply the functor $- \otimes_{\mathbb{Z}[G]} A$ to the same exact sequence and compute the homology of the resulting complex.

But for G finite, we have an isomorphism between $\mathbb{Z}[G^n] \otimes_{\mathbb{Z}[G]} A$ and $\mathrm{Hom}_{G(\mathbb{Z}[G^n],A)}$. So we can compute the Tate cohomology by applying the functor $\mathrm{Hom}_{G(-,A)}$ to the exact sequence

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z}[G^2] \longrightarrow \dots$$

where the boundary map is given by

$$\partial_n(g_1, ..., g_n) = \sum_{g\in G}\sum_{i=0}^{n} (-1)^i (g_1, ..., g_{i-1}, g, g_i, ..., g_n)$$

combining the two complexes, it is

$$\begin{array}{ccccccc} \dots & \longrightarrow & \mathbb{Z}[G^2] & \longrightarrow & \mathbb{Z}[G] & & \\ & & & \searrow & & & \\ & & \mathbb{Z}[G] & \longrightarrow & \mathbb{Z}[G^2] & \longrightarrow & \dots \end{array}$$

and the map $\mathbb{Z}[G] \to \mathbb{Z}[G]$ is given by the composition $\mathbb{Z}[G] \to \mathbb{Z} \to \mathbb{Z}[G]$, which is exactly the norm map.

We can similarly define restriction and corestriction maps for Tate cohomology.
We first define them for homology. Let $H \leq G$ be a subgroup of finite index.
Note that both $H_n(G, A)$ and $H_n(H, A|_H)$ are universal δ-functors. So we only need to define the map for $n = 0$.(need to check restriction map $A \to A|_H$ preserves projectivity and exactness)

- Corestriction: $A_H \to A_G$ is given by $a + I_H A \mapsto a + I_G A$.
- Restriction: $A_G \to A_H$ is given by $a + I_G A \mapsto \sum_{g\in H\backslash G} ga + I_H A$. (check well-defined)

For Tate cohomology, we only need to show that the restriction and corestriction maps are compatible with the norm map, and the result comes from the snake lemma.
We can check it by direct computation.

Theorem 6.3 : If I is a relatively injective G-module, then $\hat{H}^n(G, I) = 0$ for all $n \in \mathbb{Z}$.

Proof: Already know for $n > 0$. For $n \leftarrow 1$, since for G finite, induced modules are isomorphic to coinduced modules, we have I is also relatively projective. So $\hat{H}^n(G, I) = 0$ for $n \leftarrow 1$.
For $n = 0, -1$, it can be checked by direct computation for induced modules. ∎

Theorem 6.4 (restriction-corestriction): Let $H \leq G$ be a subgroup of finite index. Then

- $$\mathrm{cor}_G^H \circ \mathrm{res}_G^H = [G : H]\,\mathrm{id}_{\hat{H}^n(G,A)}$$
- $\hat{H}^n(G, A)$ is of $|G|$-torsion.
- If A is of finite type, then $\hat{H}^n(G, A)$ is finite.

Now, we know that $H_1(G, \mathbb{Z})$ (or $\hat{H}^{-2}(G, \mathbb{Z})$) is the abelianization.
What is the restriction map $\mathrm{res}_G^H : H_1(G, \mathbb{Z}) \to H_1(H, \mathbb{Z})$?
This problem is important because the abelianization of the Galois group is related to the maximal abelian

subextension.
This map is called the transfer map.

$$\begin{array}{ccccccc}
0 & \longrightarrow & H_1(G,\mathbb{Z}) & \longrightarrow & H_0(G,I_G) & \longrightarrow & 0 \\
 & & \downarrow{\scriptstyle \mathrm{res}} & & \downarrow{\scriptstyle \mathrm{res}} & & \\
0 & \longrightarrow & H_1(H,\mathbb{Z}) & \longrightarrow & H_0(H,I_G) & \longrightarrow & H_0(H,\mathbb{Z}[G]) \\
 & & \uparrow & & \uparrow & & \\
0 & \longrightarrow & H_1(H,\mathbb{Z}) & \longrightarrow & H_0(H,I_H) & \longrightarrow & 0
\end{array}$$

So we know that the transfer map is given by $H_0(G,I_G) \to H_0(H,I_H)$, which is given by $H_0(G,I_G) \to H_0(H,I_G)$ and restrict on the image of the injection $H_0(H,I_H) \to H_0(H,I_G)$.
As $H_0(H,I_G) = I_G/I_HI_G$ and the map $H_0(H,I_H) \to H_0(H,I_G)$ is given by $a + I_H^2 \mapsto a + I_HI_G$, we know that

$$I_H/I_H^2 \xrightarrow{\sim} (I_H + I_HI_G)/I_HI_G$$

Also, the restriction map $H_0(G,I_G) \to H_0(H,I_G)$ is the norm map

$$\begin{aligned} N_{H\backslash G} : I_G/I_G^2 &\to I_G/I_HI_G \\ x + I_G^2 &\mapsto \sum_{g\in H\backslash G} gx + I_HI_G \end{aligned}$$

Now, for $x \in G/[G,G]$, it corresponds to $x - 1 \in I_G/I_G^2$.

$$N_{H\backslash G}\big(x - 1 + I_G^2\big) = \sum_{g\in H\backslash G} g(x-1) + I_HI_G = \sum_{g\in H\backslash G} (gx - g) + I_HI_G$$

Take R be a set of representatives of $H \setminus G$.
For each $r \in R$, there is a unique $h_r \in H$ and $r' \in R$ s.t. $rx = h_r r'$.
Moreover, $r \mapsto r'$ is a permutation of R. So we have

$$\begin{aligned} &= \sum_{r\in R} (h_r r' - r) + I_HI_G \\ &= \sum_{r\in R} (h_r r' - r') + I_HI_G \\ &= \sum_{r\in R} (h_r - 1) + I_HI_G \end{aligned}$$

which corresponds to $\prod_{r\in R} h_r \in H/[H,H]$.
So we have the following theorem:

Theorem 6.5 : The restriction map $\mathrm{res}_G^H : H_1(G,\mathbb{Z}) \to H_1(H,\mathbb{Z})$ is given by

$$x \in G/[G,G] \mapsto \prod_{r\in R} h_r \in H/[H,H]$$

where R is a set of representatives of $H \setminus G$ and for each $r \in R$, there is a unique $h_r \in H$ and $r' \in R$ s.t. $rx = h_r r'$.

Definition Let $A \times B \to C$ be a pairing of G-modules. It induces a pairing

$$C^i(G, A) \times C^j(G, B) \to C^{i+j}(G, C)$$

given by

$$(f \smile g)(g_0, ..., g_{i+j}) = \langle f(g_0, ..., g_i), g(g_i, ..., g_{i+j}) \rangle$$

Proposition 6.6 : Let A, B, C be 3 G-modules.

1. Suppose

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

and

$$0 \longrightarrow C' \longrightarrow C \longrightarrow C'' \longrightarrow 0$$

be exact sequences of G-modules.
Suppose that we have pairings $A \times B \to C$, $A' \times B \to C'$ and $A'' \times B \to C''$ which are compatible with the exact sequences.
Then

$$(\delta a'') \smile b = \delta(a'' \smile b)$$

where a'' is a cocycle in A'' and b is a cocycle in B.

2. Suppose

$$0 \longrightarrow B' \longrightarrow B \longrightarrow B'' \longrightarrow 0$$

and

$$0 \longrightarrow C' \longrightarrow C \longrightarrow C'' \longrightarrow 0$$

be exact sequences of G-modules.
Suppose that we have three pairings $A \times B \to C$, $A \times B' \to C'$ and $A \times B'' \to C''$ which are compatible with the exact sequences.
Then

$$a \smile (\delta b'') = (-1)^p \delta(a \smile b'')$$

where p is the degree of a.

Proof: Use Leibniz rule and chase the diagram of the snake lemma:
Let α be a lifting of a''. Then $\delta a'' = [d\alpha]$ and $\delta(a'' \smile b) = [d(\alpha \smile b)]$.
Use Leibniz rule, we have $d(\alpha \smile b) = d\alpha \smile b + (-1)^p \alpha \smile db = d\alpha \smile b$.

For the second part, let β be a lifting of b''. Then $\delta b'' = [d\beta]$ and $\delta(a \smile b'') = [d(a \smile \beta)]$.
Use Leibniz rule, we have $d(a \smile \beta) = da \smile \beta + (-1)^p a \smile d\beta = (-1)^p a \smile d\beta$. ■

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Theorem 6.7 : Let G be a finite group. Then there is a unique family of pairings

$$\hat{H}^p(G, A) \times \hat{H}^q(G, B) \to \hat{H}^{p+q}(G, C)$$

associate to all $p, q \in \mathbb{Z}$ and any pairing $A \times B \to C$ of G-modules, satisfying:

- $p = q = 0$, the map are induced by the pairing $A^G \times B^G \to C^G$ given by the restriction of the pairing $A \times B \to C$.
- These maps are functorial in the category of G-modules.
- These maps are compatible with connecting homomorphisms in the sense of the previous proposition.

Proof: We want to define the pairing on the level of cochains.
We know that the Tate cohomology can be computed by the cochain complex

$$X_\bullet : \ldots \longrightarrow \mathbb{Z}[G^2] \longrightarrow \mathbb{Z}[G]$$
$$\mathbb{Z}[G] \longrightarrow \mathbb{Z}[G^2] \longrightarrow \ldots$$

apply the functor $\mathrm{Hom}_{G(-,A)}$ to the complex, we get a cochain complex computing $\hat{H}^n(G, A)$.
We want to construct $\varphi_{p,q} : X_{p+q} \to X_p \otimes_{\mathbb{Z}} X_q$ satisfying the following conditions:

- Leibniz rule: $\varphi_{p,q} \circ \partial_{p+q+1} = (\partial_{p+1} \otimes \mathrm{id}) \circ \varphi_{p+1,q} + (-1)^p (\mathrm{id} \otimes \partial_q) \circ \varphi_{p,q+1}$
- For $p = q = 0$, $\pi : \mathbb{Z}[G] \to \mathbb{Z}$, $(\pi \otimes \pi) \circ \varphi_{0,0} = \pi$.

Assume that we have constructed all $\varphi_{p,q}$, then we can complete the proof by the similar argument as the previous proposition.

We can construct $\varphi_{p,q}$ like this:

- $p, q \geq 0$:

$$\varphi_{p,q}(g_0, \ldots, g_{p+q}) = (g_0, \ldots, g_p) \otimes (g_p, \ldots, g_{p+q})$$

- $p, q \geq 1$:

$$\varphi_{-p,-q}(g_1, \ldots, g_{p+q}) = (g_1, \ldots, g_p) \otimes (g_{p+1}, \ldots, g_{p+q})$$

- $p \geq 0, q \geq 1$:

$$\varphi_{p,-p-q}(g_1, \ldots, g_q) = \sum_{(h_1, \ldots, h_p) \in G^p} (g_1, h_1, \ldots, h_p) \otimes (h_p, \ldots, h_1, g_1, \ldots, g_q)$$

Other cases are defined in the textbook. It is a long but straightforward computation to check the Leibniz rule and the condition for $p = q = 0$. ■

Proposition 6.8 :

1. If we identify $(A \otimes B) \otimes C$ with $A \otimes (B \otimes C)$, then for any $\alpha \in \hat{H}^*(G, A), \beta \in \hat{H}^*(G, B), \gamma \in \hat{H}^*(G, C)$, we have:
$$(\alpha \smile \beta) \smile \gamma = \alpha \smile (\beta \smile \gamma)$$
2. If we identify $A \otimes B$ with $B \otimes A$, then for any $\alpha \in \hat{H}^p(G, A), \beta \in \hat{H}^q(G, B)$, we have
$$\alpha \smile \beta = (-1)^{pq} \beta \smile \alpha$$
3. If $H \leq G$ is a subgroup, then restriction is compatible with the cup product, i.e. for any $\alpha \in \hat{H}^*(G, A), \beta \in \hat{H}^*(G, B)$, we have
$$\mathrm{res}^H_G(\alpha \smile \beta) = \mathrm{res}^H_G(\alpha) \smile \mathrm{res}^H_G(\beta)$$
4. If $H \lhd G$, then the inflation map is compatible with the cup product, i.e. for any $\alpha \in \hat{H}^*(G/H, A^H), \beta \in \hat{H}^*(G/H, B^H)$, we have
$$\inf_{G/H}(\alpha \smile \beta) = \inf_{G/H}(\alpha) \smile \inf_{G/H}(\beta)$$
5. If H is a subgroup of G, then the corestriction map is compatible with the cup product and the restriction map, i.e. for any $\alpha \in \hat{H}^*(H, A), \beta \in \hat{H}^*(G, B)$, we have
$$\mathrm{cor}(\alpha \smile \mathrm{res}(\beta)) = \mathrm{cor}(\alpha) \smile \beta$$

Proof: The key idea is using dimension shifting to reduce to the case of $p = q = 0$.

We only prove 2. and 5.

For 2., if $p > 0$, consider the exact sequence of G-modules
$$0 \longrightarrow A \longrightarrow I(A) \longrightarrow A_1 \longrightarrow 0$$
Then we may take $\alpha = \delta\alpha_1$ and δ is an isomorphism.

Here we need to show that the above short exact sequence splits as a sequence of abelian groups to show that
$$0 \longrightarrow A \otimes B \longrightarrow I(A) \otimes B \longrightarrow A_1 \otimes B \longrightarrow 0$$
is exact.

Here we can identify $I(A)$ with $\mathbb{Z}[G] \otimes A$. Then
$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}[G] \longrightarrow I_G \longrightarrow 0$$
splits as a sequence of abelian groups. Tensoring with A we get the desired splitting.

Use induction hypothesis we have $\alpha \smile \beta = \delta\alpha_1 \smile \beta = \delta(\alpha_1 \smile \beta) = \delta\big((-1)^{(p-1)q}\beta \smile \alpha_1\big) = (-1)^{(p-1)q}(-1)^q \beta \smile \delta\alpha_1 = (-1)^{pq}\beta \smile \alpha$.

If $p < 0$, consider the exact sequence of G-modules
$$0 \longrightarrow A_1 \longrightarrow I(A) \longrightarrow A \longrightarrow 0$$
and use the same argument as above.

Do the same for q and we reduce to the case of $p = q = 0$ and it is true by the construction of the cup product.

For 5., it is similar. We need to show that restriction and corestriction maps are compatible with the connecting homomorphisms.

■

6.3 1008

Remark: There are two important short exact sequences of G-modules: (which split as a sequence of abelian groups)

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}[G] \longrightarrow J_G \longrightarrow 0$$
$$0 \longrightarrow I_G \longrightarrow \mathbb{Z}[G] \longrightarrow \mathbb{Z} \longrightarrow 0$$

Example 6.9 : $G = \mathbb{Z}/m\mathbb{Z}$ with generator σ.
Then we have a exact sequence of G-modules:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\text{norm}} \mathbb{Z}[G] \xrightarrow{\sigma - 1} \mathbb{Z}[G] \xrightarrow{\text{aug}} \mathbb{Z} \longrightarrow 0$$

So we have $\Theta_{\mathbb{Z}} : \hat{H}^q(G, \mathbb{Z}) \cong \hat{H}^{q+2}(G, \mathbb{Z})$.
Similarly, tensoring above exact sequence (which splits as a sequence of abelian groups) with A, note that $\mathbb{Z}[G] \otimes A = I_G(A)$, which has vanishing Tate cohomology, we have $\Theta_A : \hat{H}^q(G, A) \cong \hat{H}^{q+2}(G, A)$ for any G-module A.

Now, we have $\hat{H}^0(G, \mathbb{Z}) = \mathbb{Z}/m\mathbb{Z}$. Using the isomorphism between $\hat{H}^0(G, \mathbb{Z})$ and $\hat{H}^2(G, \mathbb{Z})$, there is an element $a \in \hat{H}^2(G, \mathbb{Z})$ corresponding to $1 \in \mathbb{Z}/m\mathbb{Z}$.
We claim that Θ_A is given by cup product with a, i.e. $\Theta_A(\alpha) = \alpha \smile a$ for any $\alpha \in \hat{H}^q(G, A)$.
For any $\alpha \in \hat{H}^q(G, A)$, we have

$$\begin{aligned} \alpha \smile a &= \alpha \smile (\delta \circ \delta(1)) \\ &= (-1)^q \delta(\alpha \smile \delta(1)) \\ &= \delta(\delta(\alpha \smile 1)) \\ &= \delta \circ \delta(\alpha) \\ &= \Theta_A(\alpha) \end{aligned}$$

where the connecting homomorphisms are given by the short exact sequences given by above exact sequences of G-modules.

Lemma 6.10 : G: finite group. A, B: G-modules.
Let f be an inhomogeneous 1-cocycle of G with values in B.
Let $a \in A$ be an element with $N_G(a) = 0$.
Then f, a represent elements in $\hat{H}^1(G, B)$ and $\hat{H}^{-1}(G, A)$ respectively.
Then the cup product $a \smile f$ is represented by the inhomogeneous 0-cocycle

$$c = -\sum_{\sigma \in G} (\sigma \cdot a) \otimes f(\sigma)$$

Proof: Consider the exact sequence of G-modules

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z}[G] \xrightarrow{\rho} J_G \longrightarrow 0$$

tensoring with B, since

$$\iota_* : \hat{H}^1(G, B) \to \hat{H}^1(G, \mathbb{Z}[G] \otimes B) = 0$$

we know that there is a $\beta \in \mathbb{Z}[G] \otimes B$ s.t. $f(\sigma) = {}^{\sigma}\beta - \beta$.
Also, $\rho \circ f(\sigma) = 0$ for any $\sigma \in G$.
By the construction in snake lemma, we know that $f = \delta(\rho(\beta))$.
Then

$$\begin{aligned} a \smile f &= a \smile \delta(\rho(\beta)) \\ &= (-1)^{-1}\delta(a \smile \rho(\beta)) \end{aligned}$$

The connecting homomorphism here is given by the norm map between exact sequences of G-modules.
So we have

$$\begin{aligned} a \smile f &= -\delta(a \smile \rho(\beta)) \\ &= -N_G(a \otimes \beta) \\ &= -\sum_{\sigma \in G} (\sigma \cdot a) \otimes {}^{\sigma}\beta \\ &= -\sum_{\sigma \in G} (\sigma \cdot a) \otimes f(\sigma) - \sum_{\sigma \in G} (\sigma \cdot a) \otimes \beta \\ &= -\sum_{\sigma \in G} (\sigma \cdot a) \otimes f(\sigma) \end{aligned}$$

Remark: The cup product with a 0-cocycle is given by tensor product.

■

7 Lecture 3: The Tate-Nakayama theorem

Lemma 7.1 : Let p be a prime. Let G be a finite p-group and A be a nonzero G-module.
Suppose that A is torsion and p-primary. Then $A^G \neq 0$.

Proof: Let $a \in A$ be a nonzero element. Replace A by the submodule generated by a. Then A is a finite abelian p-group. (since A is finitely generated as an abelian group and torsion and p-primary)
Use the trick of the orbit-stabilizer theorem, we have

$$|A| = |A^G| + \sum_{[a]\text{: conjugacy class}} |[a]| = |A^G| + \sum_{[a]\text{: conjugacy class}} |G| \ / \ |G_a|$$

Then every term in the sum is divisible by p. So $|A^G|$ is also divisible by p.
As $0 \in A^G$, we have $|A^G| > 0$ which means $|A^G| \geq p$. So $A^G \neq 0$. ■

Definition For an abelian group M, we define

$$M\{p\} = \{m \in M : p^k m = 0 \text{ for some } k \geq 0\}$$

and

$$M[p] = \{m \in M : pm = 0\}$$

Lemma 7.2 : Let G be a finite group and A be a G-module.
Let $H \leq G$ be a subgroup of index m with $\gcd(m, p) = 1$.
Then

$$\text{Res} : H^i(G, A)\{p\} \to H^i(H, A)$$

is injective.

Proof: By the restriction-corestriction theorem, we have

$$\text{Cor}_G^H \circ \text{Res}_G^H = m \, \text{id}_{H^i(G,A)}$$

So if $c \in \ker(\text{Res})$, then $\text{ord}(c) \mid m$, which is coprime to p. So $c = 0$. ∎

Remark: If A is a G-module, we have two different G-actions on $I_G(A)$:

$$(\sigma f)(g) = f(\sigma^{-1} g)$$

and

$$(\sigma f)(g) = \sigma f(\sigma^{-1} g)$$

These two actions are isomorphic via the map $f \mapsto (g \mapsto gf(g))$.

Lemma 7.3 : Let G be a finite group. A, B be G-modules.
1. If B is a induced G-module, then $\text{Hom}_{\mathbb{Z}}(A, B)$ is a induced G-module.
2. If A is a projective G-mdoule. Then $A \otimes_{\mathbb{Z}} B$ is a relatively projective G-module.

Proof:
1. Use the adjunction between $\text{Hom}_{\mathbb{Z}}$ and $\otimes_{\mathbb{Z}}$.
2. A is projective $\implies$ A is a direct summand of a free G-module. Tensoring with B we get the desired result.

∎

Definition Let G be a finite group. A G-module A is called cohomologically trivial if $H^i(H, A) = 0$ for all $i \geq 1$ and all subgroups $H \leq G$.

Lemma 7.4 : Denote G_p be a p-Sylow subgroup of G.
A G-module A is G-cohomologically trivial if and only if A is G_p-cohomologically trivial for all primes p dividing $|G|$.

Proof: First note that it does not depend on the choice of G_p since any two p-Sylow subgroups are conjugate, and this gives us an isomorphism between the cohomology groups.

- $\Longrightarrow$: is obvious by definition.
- $\Longleftarrow$: Let p be a prime. Then

$$\mathrm{Res}_G^{G_p} : H^i(G, A)\{p\} \to H^i\big(G_p, A\big)$$

is injective. So if $H^i\big(G_p, A\big) = 0$, then $H^i(G, A)\{p\} = 0$.
So $H^i(G, A) = 0$ for all $i \geq 1$.
Replace G by any subgroup $H \leq G$, since the Sylow subgroups of H are subgroups of the Sylow subgroups of G, we similarly have $H^i(H, A) = 0$ for all $i \geq 1$. ∎

Theorem 7.5 : Let p be a prime. Let G be a finite p-group and A be a p-torsion G-module.
Assume that $\hat{H}^i(G, A) = 0$ for some $i \in \mathbb{Z}$.
Then A is a free $\mathbb{F}_p[G]$-module.
In particular, A is an induced module.

Proof: Since A is a p-torsion module, A is a $\mathbb{F}_p$ vector space.
So there is a $\mathbb{F}_p$ basis $\{a_i\}_{i\in I}$ of A^G.
Set $V = \bigoplus_{i\in I} \mathbb{F}_p[G]$. Then $V^G = \bigoplus_{i\in I} \mathbb{F}_p \sum_{\sigma\in G} \sigma \cong \bigoplus_{i\in I} \mathbb{F}_p$.
So we have a G-module map $\varphi : V^G \to A^G$ induced by $\mathbb{F}_p[G]^G \cong \mathbb{F}_p \cong \mathbb{F}_p a_i$.
Consider the exact sequence of $\mathbb{F}_p$-vector spaces:

$$0 \longrightarrow A^G \longrightarrow A \longrightarrow A/A^G \longrightarrow 0$$

Taking $\mathrm{Hom}_{\mathbb{F}_p}(-, V)$. We get

$$0 \longrightarrow \mathrm{Hom}_{\mathbb{F}_p}\big(A/A^G, V\big) \longrightarrow \mathrm{Hom}_{\mathbb{F}_p}(A, V) \longrightarrow \mathrm{Hom}_{\mathbb{F}_p}\big(A^G, V\big) \longrightarrow 0$$

Since $\mathrm{Hom}_{\mathbb{F}_p}(-,-)^G = \mathrm{Hom}_G(-,-)$ and using the fact that V is an induced module, we have

$$0 \longrightarrow \mathrm{Hom}_G\big(A/A^G, V\big) \longrightarrow \mathrm{Hom}_G(A, V) \longrightarrow \mathrm{Hom}_G\big(A^G, V\big) \longrightarrow 0$$

Note that φ^{-1} is an element in $\mathrm{Hom}_G\big(A^G, V\big)$. So there is a $\psi \in \mathrm{Hom}_G(A, V)$ s.t. $\psi\,|_{A^G} = \varphi^{-1}$.
Consider the exact sequence of G-modules

$$0 \longrightarrow \ker\psi \longrightarrow A \xrightarrow{\psi} V$$

applying the functor $(-)^G$, we get

$$0 \longrightarrow (\ker\psi)^G \longrightarrow A^G \xrightarrow{\psi} V^G$$

Hence $(\ker\psi)^G = 0$. By lemma 1, we have $\ker\psi = 0$. So ψ is injective.
Consider the exact sequence of G-modules

$$0 \longrightarrow A \xrightarrow{\psi} V \longrightarrow V/\psi(A) \longrightarrow 0$$

Suppose that $H^1(G, A) = 0$, then

$$0 \longrightarrow A^G \longrightarrow V^G \longrightarrow (V/\psi(A))^G \xrightarrow{\delta} H^1(G, A) \longrightarrow 0$$

hence $(V/\psi(A))^G = 0$ and then $V/\psi(A) = 0$. So ψ is an isomorphism.
So it suffices to show that $H^1(G, A) = 0$.
If $i = 1$, we are done.
Using dimension shifting, we get a p-torsion G-module A' s.t. $\hat{H}^n(G, A) = \hat{H}^{n-i+1}(G, A')$.
Then $\hat{H}^1(G, A') = 0$, so A' is a free $\mathbb{F}_p[G]$-module. In particular, $\hat{H}^n(G, A') = 0$ for all $n \in \mathbb{Z}$. So $\hat{H}^1(G, A) = 0$. ∎

Theorem 7.6 : Let G be a finite group and A be a G-module.

1. Assume that for any prime p, there is some $i \in \mathbb{Z}$ s.t. $\hat{H}^i(G_p, A) = \hat{H}^{i+1}(G_p, A) = 0$, where G_p is a p-Sylow subgroup of G. Then A is a cohomologically trivial G-module.
   Suppose that moreover A is a free $\mathbb{Z}$-module. Then A is a projective G-module.

Proof: Take a free G-module F and a surjection $F \to A$. Let $R = \ker(F \to A)$. Then we have an exact sequence of G-modules

$$0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0$$

As F is free, we have R is a free $\mathbb{Z}$-module.
Then

$$\hat{H}^{i+1}(G_p, R) = \hat{H}^{i+2}(G_p, R) = 0$$

Also, if R is cohomologically trivial, then A is cohomologically trivial. So we can replace A by R.
Now we assume that A is a free $\mathbb{Z}$-module.
Consider the exact sequence

$$0 \longrightarrow A \xrightarrow{\mathrm{p}} A \longrightarrow A/pA \longrightarrow 0$$

Then $\hat{H}^i(G_p, A/pA) = 0$. By the previous theorem, A/pA is a free $\mathbb{F}_p[G_p]$-module.
Taking $\mathrm{Hom}_{\mathbb{Z}}(A, -)$, as A is a free $\mathbb{Z}$-module, we have

$$0 \longrightarrow M := \mathrm{Hom}_{\mathbb{Z}}(A, R) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(A, F) \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(A, A) \longrightarrow 0$$

If $H^1(G, M) = 0$, then

$$0 \longrightarrow M^G \longrightarrow \mathrm{Hom}_G(A, F) \longrightarrow \mathrm{Hom}_G(A, A) \longrightarrow 0$$

Since id_A is an element of $\mathrm{Hom}_G(A, A)$, there is a G-lifting of $F \to A$. So A is a direct summand of F.
Hence A is a projective G-module. In particular, A is cohomologically trivial.

So it suffices to show that $H^1(G, M) = 0$.
We have the exact sequence of G-modules

$$0 \longrightarrow M \xrightarrow{\mathrm{p}} M \longrightarrow \mathrm{Hom}_{\mathbb{Z}}(A, R/pR) \longrightarrow 0$$

By the previous theorem, R/pR is a free $\mathbb{F}_p[G_p]$-module. So $M/pM = \mathrm{Hom}_{\mathbb{Z}}(A, R/pR)$ is an induced G_p-module.
In particular, we have

$$0 \longrightarrow H^1(G_p, M) \xrightarrow{\mathrm{p}} H^1(G_p, M) \longrightarrow 0$$

So $H^1(G_p, M)[p] = 0$.
By lemma 2, we have $H^1(G, M)\{p\} = 0$.
As p is arbitrary, we have $H^1(G, M) = 0$. ■

7.1 1015

Theorem 7.7 : Assume that A is cohomologically trivial. Then $\hat{H}^n(H, A) = 0$ for any subgroup $H \leq G$ and any $n \in \mathbb{Z}$ and there's an exact sequence of G-modules

$$0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0$$

where F is a free G-module and R is a projective G-module.

Theorem 7.8 : Let A be a cohomologically trivial G-module. Let B be a $\mathbb{Z}[G]$-module which is free as a $\mathbb{Z}$-module.
Then $A \otimes_{\mathbb{Z}} B$ is a cohomologically trivial G-module.

Proof: We have an exact sequence of G-modules

$$0 \longrightarrow R \longrightarrow F \longrightarrow A \longrightarrow 0$$

where F is a free G-module and R is a projective G-module.
Since B is free as a $\mathbb{Z}$-module, we have an exact sequence of G-modules

$$0 \longrightarrow R \otimes_{\mathbb{Z}} B \longrightarrow F \otimes_{\mathbb{Z}} B \longrightarrow A \otimes_{\mathbb{Z}} B \longrightarrow 0$$

Since $F \otimes_{\mathbb{Z}} B$ is a coinduced H-module and $R \otimes_{\mathbb{Z}} B$ is a relatively projective H-module, we have $\hat{H}^n(H, A \otimes_{\mathbb{Z}} B) = 0$ for all $n \in \mathbb{Z}$. ■

Proposition 7.9 : Let A, A' be G-modules and $f : A' \to A$ be a G-module homomorphism.
Assume for each prime p, there is an $n_p \in \mathbb{Z}$ s.t. $f_* : \hat{H}^i(G_p, A') \to \hat{H}^i(G_p, A)$ is surjective for $i = n_p$, bijective for $i = n_p + 1$ and injective for $i = n_p + 2$.
Then for every subgroup $H \leq G$ and any G-module B which is free as a $\mathbb{Z}$-module, the map

$$\hat{H}^i(H, A' \otimes_{\mathbb{Z}} B) \to \hat{H}^i(H, A \otimes_{\mathbb{Z}} B)$$

is isomorphism.

Proof:

- Case 1. f is injective. Then we have an exact sequence of G-modules

$$0 \longrightarrow A' \xrightarrow{f} A \longrightarrow A'' \longrightarrow 0$$

  By assumption, we have $\hat{H}^i(G_p, A'') = 0$ for $i = n_p, n_p + 1$.
  By the previous theorem, A'' is a cohomologically trivial G-module.
  By the previous theorem, $A'' \otimes_{\mathbb{Z}} B$ is a cohomologically trivial G-module.
  In particular, $\hat{H}^i(H, A' \otimes_{\mathbb{Z}} B) \cong \hat{H}^i(H, A \otimes_{\mathbb{Z}} B)$ for all $i \in \mathbb{Z}$.
- In general, replace A by $A \oplus I_G(A')$, then $f \oplus \iota$ becomes injective and $I_G(A')$ does not contribute to the Tate cohomology. So we reduce to the previous case.

■

Proposition 7.10 : Let $A \times B \to C$ be a pairing of G-modules.
Take $[a] \in \hat{H}^q(G, A)$. Denote $[a_H]$ the restriction of $[a]$ to $\hat{H}^q(H, A)$.
For any $n \in \mathbb{Z}$, G-module D, and any subgroup $H \leq G$, we consider

$$f(n, H, D) : \hat{H}^n(H, B \otimes_{\mathbb{Z}} D) \to \hat{H}^{n+q}(H, C \otimes_{\mathbb{Z}} D)$$

given by $[b] \mapsto [a_H] \smile [b]$.
Assume that for each prime p, there is an $n_p \in \mathbb{Z}$ s.t. $f(n_p, G_p, \mathbb{Z})$ is surjective, $f(n_p + 1, G_p, \mathbb{Z})$ is bijective, and $f(n_p + 2, G_p, \mathbb{Z})$ is injective.
Then for every subgroup $H \leq G$ and any G-module D which is free as a $\mathbb{Z}$-module, the map

$$f(n, H, D) : \hat{H}^n(H, B \otimes_{\mathbb{Z}} D) \to \hat{H}^{n+q}(H, C \otimes_{\mathbb{Z}} D)$$

is an isomorphism.

Proof: If $q = 0$, then we are done by the previous proposition.
To use dimension shifting, we will use the following two exact sequences of G-modules:

$$0 \longrightarrow A \longrightarrow I_G(A) \longrightarrow A_1 \longrightarrow 0$$
$$0 \longrightarrow C \longrightarrow I_G(C) \longrightarrow C_1 \longrightarrow 0$$

which is compatible with the pairing $A \times B \to C$.
Then $f(n, H, D)$ and $f_1(n, H, D)$ differ by isomorphisms.(since D is $\mathbb{Z}$-free) So the conditions still hold and we can reduce to the case of $q = 0$.
(If $q < 0$ use another exact sequence of G-modules) ■

Theorem 7.11 (Tate-Nakayama): Let A be a G-module. $a \in \hat{H}^2(G, A)$.
Assume that for any prime p, the following holds:

- $H^1\big(G_p, A\big) = 0$.
- $H^2\big(G_p, A\big)$ is of cardinality $|G_p|$ and is generated by $\mathrm{Res}(a)$.

Then for any G-module D which is free as a $\mathbb{Z}$-module, the cup product with a induces an isomorphism

$$\hat{H}^n(H, D) \cong \hat{H}^{n+2}(H, A \otimes_{\mathbb{Z}} D)$$

Proof: Let $B = \mathbb{Z}$, $C = A$ and $n_p = -1$. ∎

8 Cohomology for profinite groups

From now on, G is a profinite group.
For a G-module A, equip A with the discrete topology.
We say that G acts continuously on A if $G \times A \to A$ is continuous.
Equivalently, for any $a \in A$, the stabilizer G_a is an open subgroup of G.
In this case, we say that A is a discrete G-module.
Denote C_G the category of discrete G-modules.

Suppose that A is a discrete G-module which is finitely generated as an abelian group. Then

$$\{g \in G : g \cdot a = a \forall a \in A\}$$

is an open normal subgroup of G. So the action of G on A factors through a finite quotient of G.

Example 8.1 :

$$\mu_n = \{z \in k^s : z^n = 1\}$$

is a discrete G_k-module.

Question: How to define the cohomology groups for profinite groups?

We can't use $\mathbb{Z}[G]$ here since it is not a discrete G-module.
In fact, if $|G|$ is infinite, C_G does not have enough projective objects.
On the other hand, C_G has enough injective objects.
So we can define the cohomology groups for profinite groups by the right derived functors of the left exact functor $A \mapsto A^G$.

Remark: C_G has enough injective objects.

Proof: We can use the Frobenius reciprocity for profinite groups.
In fact, the smooth induction functor is the smoothing of the induction functor for finite groups.

Since smoothing functor $\mathrm{Mod}_G \to C_G$ is left exact and $\mathrm{Hom}_G(V,W) = \mathrm{Hom}_{C_G}(V, W^\infty)$ if V is smooth, we have smoothing functor preserves injective objects.

In the following, we will define the cohomology groups for profinite groups using continuous cochains.

Definition Let G be a profinite group and A be a discrete G-module.
For any $n \geq 0$, we define

$$C^n(G,A) = \{f : G^n \to A : f \text{ is continuous}\}$$

with the convention that $C^0(G,A) = A$.
Define the differential

$$d^n : C^n(G,A) \to C^{n+1}(G,A)$$

by

$$d^n f(g_1, ..., g_{n+1}) = g_1 \cdot f(g_2, ..., g_{n+1}) + \sum_{i=1}^{n} (-1)^i f(g_1, ..., g_i g_{i+1}, ..., g_{n+1}) + (-1)^{n+1} f(g_1, ..., g_n)$$

Proposition 8.2 : Let Λ be a filtered index set.
Let $(G_\lambda)_{\lambda\in\Lambda}$ be a projective system of profinite groups.
Let $(A_\lambda)_{\lambda\in\Lambda}$ be a inductive system of discrete G_λ-modules.
Suppose that the action of G_λ on A_λ is compatible with the transition maps.
Define

$$G = \varprojlim_{\lambda\in\Lambda} G_\lambda$$

and

$$A = \varinjlim_{\lambda\in\Lambda} A_\lambda$$

Then for any $n \geq 0$, we have

$$H^n(G,A) \cong \varinjlim_{\lambda\in\Lambda} H^n(G_\lambda, A_\lambda)$$

Proof: First, G is profinite and A is a discrete G-module.
Fact: taking inductive limit with filtered index set is exact.
We know that the cohomology is given by the following two exact sequences of complexes:

$$0 \longrightarrow Z^\bullet \longrightarrow C^\bullet \longrightarrow B[+1]^\bullet \longrightarrow 0$$

and

$$0 \longrightarrow B^\bullet \longrightarrow Z^\bullet \longrightarrow H^\bullet \longrightarrow 0$$

So it suffices to show that

$$C^n(G,A) \cong \varinjlim_{\lambda\in\Lambda} C^n(G_\lambda, A_\lambda)$$

For any $f \in C^n(G,A)$, since G^n is compact and A is discrete, the image of f is finite.
So the image of f is contained in some A_λ.
So there are finitely many open sets V_i of G^n s.t. f is constant on each V_i.
Since G^n is compact, we may take each V_i to be of the form $V_i = \prod_{j=1}^n U_j$, where each U_j is an open set of G of the form $U_j = \pi_k^{-1}(W_k)$ for some $k \in \Lambda$ and some open set W_k of G_k.
Then we can find a $\lambda_0 \in \Lambda$ s.t. f factors through $G_{\lambda_0}^n$.
We need to show that f can extend to a continuous map $G_{\lambda_0}^n \to A_{\lambda_0}$.
Since $\pi_{\lambda_0}(G)$ is closed in G_{λ_0}, by exercise, we have a continuous section $s : G_{\lambda_0}/\pi_{\lambda_0}(G) \to G_{\lambda_0}$, which gives us a homeomorphism $G_{\lambda_0}/\pi_{\lambda_0}(G) \times \pi_{\lambda_0}(G) \cong G_{\lambda_0}$.
We already know that f can be regarded as a continuous map $\pi_{\lambda_0}(G)^n \to A_{\lambda_0}$.
So we can extend f to a continuous map $G_{\lambda_0^n} \to A_{\lambda_0}$ by composing with the projection $G_{\lambda_0^n} \to \pi_{\lambda_0}(G)^n$.

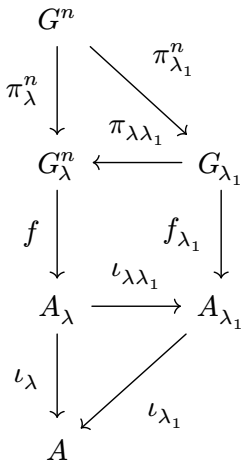
■

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continue the proof of the previous proposition.
The rest part is to show that the map

$$\varinjlim_{\lambda\in\Lambda} C^n(G_\lambda, A_\lambda) \to C^n(G,A)$$

is injective.
Each $C^n(G_\lambda, A_\lambda) \to C^n(G,A)$ is given by $f \mapsto \iota_\lambda \circ f \circ \pi_\lambda^n$ where $\pi_\lambda : G \to G_\lambda$ and $\iota_\lambda : A_\lambda \to A$ are given by the projective and inductive systems.
We need to show that if $f \in C^n(G_\lambda, A_\lambda)$ is such that $\iota_\lambda \circ f \circ \pi_\lambda^n = 0$, then there exists a $\lambda' \geq \lambda$ s.t. f factors through $G_{\lambda'}^n$ and the induced map $G_{\lambda'}^n \to A_{\lambda'}$ is zero.
Since f is continuous, the image of f is finite. So there is a $\lambda_1 \geq \lambda$ the following diagram commutes:



with $G^n \to A_{\lambda_1}$ is already zero.
Now, for $i \geq \lambda_1$, define $E_i = \left\{x \in G_i^n : f_{\lambda_1} \circ \pi_{\lambda_1 i}(x) \neq 0\right\}$.

Then $\{E_j\}$ is a projective system.
As f only take finitely many values, each E_i is a finite union of clopen sets, hence E_i is compact.
If each E_i is nonempty, since the projective limit of nonempty T2 compact spaces is nonempty, we have $\varprojlim_{i \geq \lambda_1} E_i \neq \varnothing$, which is a contradiction.
So there is a $\lambda' \geq \lambda_1$ s.t. $E_{\lambda'} = \varnothing$, which means that $f_{\lambda'} = 0$.

Corollary 8.3 : Let G be a profinite group and A be a discrete G-module.
Then

$$G = \varprojlim_{U \lhd G \text{: open}} G/U$$

and

$$A = \varinjlim_{U \lhd G \text{: open}} A^U$$

So

$$H^n(G, A) \cong \varinjlim_{U \lhd G \text{: open}} H^n(G/U, A^U)$$

which is an inductive limit of finite group cohomology groups.

Remark:

1. $H \leq G$: closed subgroup. Then we have the restriction map
$$\mathrm{Res} : H^n(G, A) \to H^n(H, A)$$
which is compatible with the restriction map for finite groups.
2. For corestriction, since the original coinduced module is not discrete, we need a new proof of the restriction of injective modules in C_G in C_H is injective.
Also, the 0-degree corestriction is defined by summing over the cosets of H in G.
So we need H to be open in G to make sure that the sum is finite.
3. Induced modules are defined before which works well for profinite groups.
4. The cohomology groups defined above is right derived functors of the left exact functor $A \mapsto A^G$.
We need to show that $H^0(G, A) = A^G$ and $H^n(G, -)$ is effaceable.
So we only need to show that $H^n(G, I)$ is zero for all injective modules I and $n \geq 1$.
By the proposition above, we have
$$H^n(G, I) \cong \varinjlim_{U \lhd G\text{: open}} H^n(G/U, I^U)$$
We claim that I^U is an injective G/U-module. (and then we are done)
From the adjunction
$$\mathrm{Hom}_G(M, I) \cong \mathrm{Hom}_{G/U}(M, I^U)$$
We have the desired result.

Exercise 8.4 : We have seen that if I is an injective G-module, then I^U is an injective G/U-module. In fact, the converse is also true. If I^U is an injective G/U-module for all open normal subgroups $U \leq G$, then I is an injective G-module.

Proposition 8.5 : If I is injective in C_G, then I is injective in C_H for any closed subgroup $H \leq G$.

Proof: It suffices to show that I^V is injective for any open normal subgroup $V \leq H$.
Let V be an open normal subgroup of H.
For $U \cap H \leq V$, we have I^U is injective in $C_{G/U}$.
Then
$$H/U \cap H \leq G/U$$
So I^U is injective in $C_{H/U \cap H}$.
Consider UV, which is an open normal subgroup of G.
Then
$$I^V = \varinjlim_{U\text{: open normal subgroup of G}} I^{UV}$$
So we only need to show that I^{UV} is injective in $C_{H/V}$ by the following exercise.
We have I^{UV} is injective in $C_{G/UV}$.

Also, $G/UV \geq H/(UV \cap H) = H/V$.
So we are done. ■

Exercise 8.6 : Let R be a ring which is left Noetherian. Let $\{M_i\}$ be a filtered inductive system of injective R-modules. Then $\varinjlim_i M_i$ is an injective R-module.

Hint: Use Baer's criterion.